

Big, Affordable, and Green

1 400 on Fifth, a condominium building located in Harlem, New York, celebrated its grand opening in October, 2004, marking its status as the largest affordable green multifamily residence in the country. Designed by the firm Full Spectrum, which specializes in developing affordable green and smart buildings for urban markets, the \$40 million, 225,000 ft² building consists of 129 condominium homes, 9 townhouses, and first-floor retail space. Two-thirds of the condominiums are reserved for buyers with an annual income of \$50,000–\$104,000, while the last third are selling at market rates. Retail residents include a sports club, a dry cleaner, a bank, and a restaurant. Housing Commissioner Shaun Donovan says that “1400 on Fifth gives 129 families the opportunity to be homeowners and to build their futures in Harlem.”

1400 on Fifth is the first affordable urban housing development to receive a New Construction Whole Building Award for its energy design. It is also expected to qualify for the New York State Green Building tax credit, as well as a U.S. Green Building Council LEED Award. Steven Winter Associates, Incorporated, oversaw the project's design to ensure that these standards could be met.

The building uses 35% less energy than permitted by the New York State Energy Code, and 70% less energy than similar buildings in New York City. IAQ is high, thanks to filtered fresh air that removes allergens and eliminates materials that off-gas. Over 60% of the building is constructed from recycled or renewable resources. The building's heating and cooling system is geothermal, which prevents the production of 700 tons of CO₂ annually. The use of geothermal power also eliminates dependence on fossil fuels for heating and cooling. Devashish Lahiri, a senior associate with Steven Winter Associates, notes that “the team had a commitment to the energy efficient system. New York State incentives, and a favorable aquifer/water source, helped to make



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that decision. The heat pumps are expected to operate at a 20+ energy efficiency ratio (EER) at Air Conditioning & Refrigeration Institute (ARI) conditions and will reduce fossil fuel dependency in New York.”

Lahiri also noted that a number of other products and techniques also raise the building's energy efficiency. These include “geothermal heat pumps and GFX [a graywater heat recovery system] for service hot water heat recovery. In addition, a low-e glazing system; variable-frequency drive on pumps; premium motors; and many commonsense items, such as LED exit signs and CFLs were used. In aggregate, these will reduce carbon emissions by approximately 300 tons per year.”

The land for 1400 on Fifth was provided by the NYC Department of Housing Preservation and Development, which also initiated rezoning and provided key funding for the project.

Additional funding and financing was provided by Bank of America, Fannie Mae, the Housing Partnership Development Corporation, the NYC Housing Development Corporation, HUD, Empire State Development Corporation, and the New York State Energy Research and Development Authority.



—Elka Karl

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For More Information:

To learn more about 1400 on Fifth, go to www.1400on5th.com.